

Lattice non-perturbative understanding of generalized symmetries and applications

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- M. Abe, OM and S. Onoda, PRD **108**, 014506 (2023) [2304.11813].
- Y. Honda, OM, S. Onoda and H. Suzuki, PTEP **2024**, no.4, 043B04 (2024) [2401.01331].

Acknowledgements

- At first studied generalized symmetry with Hiroki Wada and Satoshi Yamaguchi [PRD **107**, 045020 (2023)]
- Lattice understanding (Abelian gauge theory)
 - ▶ Fractional topo charge with Motokazu Abe and Hiroshi Suzuki [PTEP **2023**, no.2, 023B03 (2023)]
 - ▶ Higher group with Naoto Kan, Yuta Nagoya and Hiroki Wada [EPJC **83**, no.6, 481 (2023)]
- Non-Abelian generalization with Motokazu Abe, Soma Onoda, Hiroshi Suzuki and Yuya Tanizaki [JHEP **08**, 118 (2023)]

Symmetry and 't Hooft anomaly matching

- **Symmetry**: fundamental tool in physics
- Noether theorem and conservation law

$$\text{Sym} : \phi \mapsto \phi'; S(\phi) = S(\phi'), \quad \partial_\mu j^\mu = 0$$

$$\text{Charge } Q \equiv \int j^0 d^{D-1}x$$

- **'t Hooft anomaly matching** for strongly coupled theories [79]
 - ▶ Assume global symmetry G in system
 - ▶ Introduce background gauge field A assoc. G (by gauging G)

$$\mathcal{Z}[A] = \int \mathcal{D}\{\phi\} e^{-S(\{\phi\}, A)} \stackrel{?}{=} \mathcal{Z}[A^g] = \dots = e^{\mathcal{A}[A, g]} \mathcal{Z}[A]$$

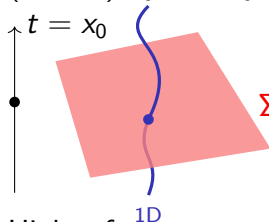
$e^{\mathcal{A}} \neq 1$ **Anomalous** 't Hooft anomaly which is invariant at any energy scale (**renormalization group inv.**)

- ▶ Restriction on low-energy dynamics: **SSB, phase structure, SPT**

Generalization: higher-form symmetry

- Generalized global symmetry [Gaiotto–Kapustin–Seiberg–Willet '14]
 - Coupled with topological field theory [Kapustin–Seiberg '14]

- (0-form) Symmetry
 - Charge (codim 1)



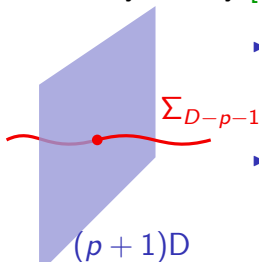
$$Q \equiv \int_{\Sigma_{D-1}} j_0 dx_1 \wedge \cdots \wedge dx_{D-1}$$

Σ_{D-1} Symmetry operator

$$U_\alpha(\Sigma_{D-1}) = e^{i\alpha Q}$$

Topological under deformation of Σ_{D-1}

- Higher-form symmetry [Gaiotto–Kapustin–Seiberg–Willet '14]



- p -form symmetry $G^{[p]}$ (codim $p+1$)

$$Q \equiv \int_{\Sigma_{D-p-1}} \star j^{(p+1)}, \quad U_\alpha(\Sigma) = e^{i\alpha Q}$$

- Transforming a “loop” operator $W(C)$
 $W(C) \mapsto U(\Sigma)W(C) = e^{i\alpha \#(\Sigma, C)} W(C)$

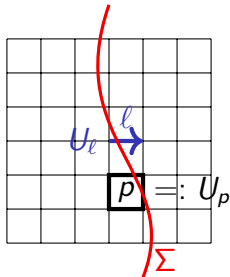
Center symmetry in YM theory

- Lattice $SU(N)$ YM theory
 - ▶ link variable $U_\ell \in SU(N)$

- Center symmetry: $\mathbb{Z}_N^{[1]}$

$$e^{\frac{2\pi i}{N}k} \in \mathbb{Z}_N \subset SU(N); \quad U_\ell \mapsto e^{\frac{2\pi i}{N}k \#(\Sigma, \ell)} U_\ell$$

Intersection $\#$ of Σ & link ℓ ; $U_p \mapsto U_p$

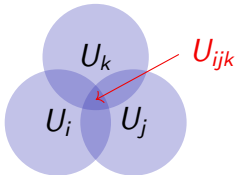


- Gauging the center symmetry

$$S \sim \sum \text{Tr} e^{-\frac{2\pi i}{N} B_p} U_p \quad B_p: \text{2-form gauge field assoc. } \mathbb{Z}_N^{[1]}$$

invariant under $U_\ell \mapsto e^{\frac{2\pi i}{N}\lambda_\ell} U_\ell$, $B_p \mapsto B_p + (d\lambda)_p$

- Consistency condition for transition function g_{ij} :



Cocycle condition: $g_{ij}g_{jk}g_{ki} = 1$ at U_{ijk}
 $\rightarrow g_{ij}g_{jk}g_{ki} \in \mathbb{Z}_N$ under B_p background.

Our enterprise: Fully lattice regularized framework

- Original observation

- ▶ $Q = \frac{1}{8\pi^2} \int F \wedge F \in \mathbb{Z}$
- ▶ Gauging $\mathbb{Z}_N$: $Q = -\frac{\varepsilon_{\mu\nu\rho\sigma} Z_{\mu\nu} Z_{\rho\sigma}}{8N} + \mathbb{Z}$
- ▶ Indicating mixed 't Hooft anomaly with chiral sym/ θ -periodicity

$$\mathcal{Z}_{\theta+2\pi}[B_p] = e^{-2\pi i Q} \mathcal{Z}_{\theta}[B_p] \quad \text{not } 2\pi \text{ periodic}$$

★ $\theta = 0$ (monopole condensation) $\leftrightarrow \theta = 2\pi$ (dyon condensation)

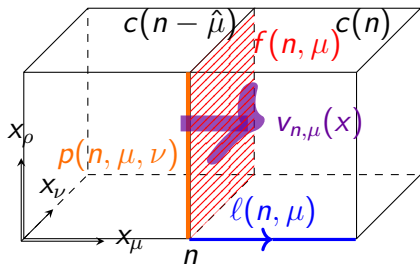
- Fully regularized framework: Lattice regularization

- ▶ Use lattice construction of fiber bundle [Lüscher '84]
- ▶ Prove $Q \in \frac{1}{N}\mathbb{Z}$ [Abe-OM-Onoda-Suzuki-Tanizaki]

- Application to higher-group, non-invertible symmetry

Bundle structure on lattice?

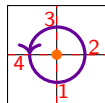
- No continuity for lattice fields?
 - ▶ Any configuration can be deformed continuously to others
 - ▶ We can observe topological structure even on lattice by Lüscher
- Setup/strategy: Lattice Λ divides X into 4D hypercubes



4D cell $c(n)$
 3D face $f(n, \mu)$
 2D plaquette $p(n, \mu, \nu)$
 1D link $\ell(n, \mu)$

- 1 Regard $\{c(n)\}$ as patches
- 2 Define transition function $v_{n, \mu}(x)$ at $f(n, \mu)$ from data as U_ℓ

- ▶ Difficult to define it at $x \neq n$ s.t.
cocycle condition is kept intact



$$v_1 v_2 v_3^{-1} v_4^{-1} = 1$$

at $x \in p(n)$

Fractionality of topology on lattice

- Lüscher's construction
Coupled with higher-form gauge fields

$$v_{n,\mu}(x) = S_{n,\mu}^{n-\hat{\mu}}(x)^{-1} U_{n-\hat{\mu},\mu} S_{n,\mu}^n(x)$$

$$S \equiv S[U_p]$$

- Replacing $U_p \rightarrow \tilde{U}_p$,
where $\tilde{U}_p = e^{-\frac{2\pi i}{N} B_p} U_p$

$$v \rightarrow \tilde{v}, S \rightarrow \tilde{S} \equiv \tilde{S}[\tilde{U}_p]$$

- Everything becomes $\mathbb{Z}_N$ 1-form gauge invariant!
- Cocycle condition can take a $\mathbb{Z}_N$ value:

$$\tilde{v}_{n-\hat{\nu},\mu}(x) \tilde{v}_{n,\nu}(x) \tilde{v}_{n,\mu}(x)^{-1} \tilde{v}_{n-\hat{\mu},\nu}(x)^{-1} = e^{\frac{2\pi i}{N} B_{\mu\nu}(n-\hat{\mu}-\hat{\nu})}$$

$$Q \in -\frac{\varepsilon_{\mu\nu\rho\sigma} Z_{\mu\nu} Z_{\rho\sigma}}{8N} + \mathbb{Z}$$

- We observe mixed 't Hooft anomaly with lattice regularization!

Construction of topo. sectors on lattice [Lüscher]

- Parallel transporter (interpolation):

$$U_\ell \rightarrow v_f(n) \rightarrow v_f(x) \leftarrow \text{Cocycle}$$

$$v_{n,\mu}(n) = U(n - \hat{\mu}, \mu)$$

$$v_{n,\mu}(x) \equiv S_{n,\mu}^{n-\hat{\mu}}(x)^{-1} v_{n,\mu}(n) S_{n,\mu}^n(x)$$

$$w^n(x) = U_{s_1, s_2}^{n-\hat{s}_1} U_{s_2, s_3}^{n-\hat{s}_2} U_{s_3, s_4}^{n-\hat{s}_3} U_{s_4, s_1}^{n-\hat{s}_4}$$

$$f_{s,\mu}^m(x_\gamma) = (u_{s_2 s_3}^m)^{\gamma_\gamma} (u_{s_3 s_4}^m)^{\gamma_\gamma} (u_{s_4 s_1}^m)^{\gamma_\gamma} (u_{s_1 s_2}^m)^{\gamma_\gamma} (u_{s_2 s_3}^m)^{\gamma_\gamma} (u_{s_3 s_4}^m)^{\gamma_\gamma} (u_{s_4 s_1}^m)^{\gamma_\gamma} (u_{s_1 s_2}^m)^{\gamma_\gamma}$$

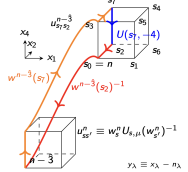
$$g_{s,\mu}^m(x_\gamma) = (u_{s_2 s_3}^m)^{\gamma_\gamma} (u_{s_3 s_4}^m)^{\gamma_\gamma} (u_{s_4 s_1}^m)^{\gamma_\gamma} (u_{s_1 s_2}^m)^{\gamma_\gamma} (u_{s_2 s_3}^m)^{\gamma_\gamma} (u_{s_3 s_4}^m)^{\gamma_\gamma} (u_{s_4 s_1}^m)^{\gamma_\gamma} (u_{s_1 s_2}^m)^{\gamma_\gamma}$$

$$h_{s,\mu}^m(x_\gamma) = (u_{s_2 s_3}^m)^{\gamma_\gamma} (u_{s_3 s_4}^m)^{\gamma_\gamma} (u_{s_4 s_1}^m)^{\gamma_\gamma} (u_{s_1 s_2}^m)^{\gamma_\gamma} (u_{s_2 s_3}^m)^{\gamma_\gamma} (u_{s_3 s_4}^m)^{\gamma_\gamma} (u_{s_4 s_1}^m)^{\gamma_\gamma} (u_{s_1 s_2}^m)^{\gamma_\gamma}$$

$$k_{s,\mu}^m(x_\gamma) = (u_{s_2 s_3}^m)^{\gamma_\gamma} (u_{s_3 s_4}^m)^{\gamma_\gamma} (u_{s_4 s_1}^m)^{\gamma_\gamma} (u_{s_1 s_2}^m)^{\gamma_\gamma} (u_{s_2 s_3}^m)^{\gamma_\gamma} (u_{s_3 s_4}^m)^{\gamma_\gamma} (u_{s_4 s_1}^m)^{\gamma_\gamma} (u_{s_1 s_2}^m)^{\gamma_\gamma}$$

$$l_{s,\mu}^m(x_\beta, x_\gamma) = [f_{s,\mu}^m(x_\gamma)]^{-1} [f_{s,\mu}^m(x_\gamma) k_{s,\mu}^m(x_\gamma) g_{s,\mu}^m(x_\gamma)]^{-1} h_{s,\mu}^m(x_\gamma)^{-1} [f_{s,\mu}^m(x_\gamma)]^{\gamma_\beta} h_{s,\mu}^m(x_\gamma) [g_{s,\mu}^m(x_\gamma)]^{\gamma_\beta}$$

$$S_{n,\mu}^m(x_\alpha, x_\beta, x_\gamma) = (u_{s_2 s_3}^m)^{\gamma_\gamma} [f_{s,\mu}^m(x_\gamma)]^{\gamma_\beta} [g_{s,\mu}^m(x_\beta, x_\gamma)]^{\gamma_\alpha}$$



- Topo. sectors on lattice so that $[Q \text{ in terms of } v_f(x)] \in \mathbb{Z}$

$$Q = \sum_{n \in \Lambda} q(n), \quad q(n) = -\frac{1}{24\pi^2} \sum_{\mu, \nu, \rho, \sigma} \varepsilon_{\mu\nu\rho\sigma} \left\{ 3 \int_{p_n, \mu\nu} d^2x \text{Tr} \left[(v_{n,\mu} \partial_\rho v_{n,\mu}^{-1}) (v_{n-\hat{\mu}, \nu} \partial_\sigma v_{n-\hat{\mu}, \nu}^{-1}) \right] \right. \\ \left. + \int_{f_{n,\mu}} d^3x \text{Tr} \left[(v_{n,\mu}^{-1} \partial_\nu v_{n,\mu}) (v_{n,\mu}^{-1} \partial_\rho v_{n,\mu}) (v_{n-\hat{\mu}, \nu}^{-1} \partial_\sigma v_{n,\mu}) \right] \right\}$$

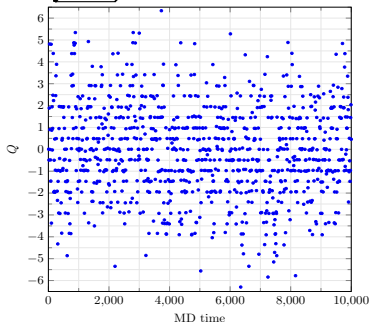
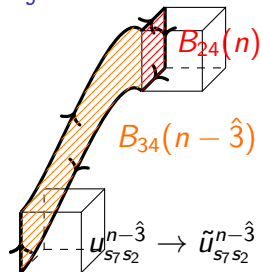
Numerical simulations of higher-form gauge fields

<https://github.com/o-morikawa/Gaugefields.jl>

- Obviously, we can implement it on computer by $\forall \text{loop } u \rightarrow \tilde{u}$
- External/dynamical B to carry out MC, gradient flow...
 - ▶ Numerically fractional topo. charge is shown

Recent studies:

- Reduction of autocorrelation in $SU(N)/\mathbb{Z}_N$ MC [Abe–OM–Suzuki '25]
- 't Hooft partition function (free energy w/ flat B) [OM–Suzuki '25, wip]
- Wilson–'t Hooft classification w/ nonflat B [on-going Kanamori–OM–Tanizaki–Tomiya–Watanabe]



Axial $U(1)$ non-invertible symmetry in continuum

- Quantum anomaly $\rightarrow$ loss of symmetry
ABJ anomaly: no conserved current for axial $U(1)$ symmetry
- For fractional rotation angles, conserved, gauge-invariant and topological **non-invertible** symmetry exists

[Córdova–Ohmori '22, Choi–Lam–Shao '22]

[Many studies; See recent lectures, reviews & PhD theses in [gsym.rev.bib](#)]

- Naive symmetry operator

$$d \star j = \frac{1}{4\pi^2} F \wedge F, \quad U_\alpha(M) = \exp \left(\frac{i\alpha}{2} \int_M \star j \right)$$

is not topological

- Let us consider topological modification by Chern–Simons:

$$\hat{U}_\alpha(M) = \exp \left[\frac{i\alpha}{2} \int \left(\star j - \frac{1}{4\pi^2} A \wedge dA \right) \right]$$

But gauge non-invariant!!

- For $\alpha = \frac{2\pi}{N}$, $-i \int_M \frac{1}{4\pi N} A \wedge dA$ is still gauge non-invariant because of fractional CS level $1/N$

Fractional quantum Hall effect and non-invertibility

- TQFT (fractional quantum Hall state) as

$$i \int_M \left(\frac{N}{4\pi} a \wedge da + \frac{1}{2\pi} a \wedge dA \right)$$

which is gauge-invariant with fractional CS level

- ▶ a : additional dynamical $U(1)$ gauge field
- ▶ Naively, this action is (level- N CS) + $(A_\mu J^\mu)$; $a = -A/N$
“Substituting” this into action reproduces $-i \int \frac{1}{4\pi N} A \wedge dA$
- To modify naive symmetry operator $U_\alpha(M)$, we use this action at boundary M instead of naive CS term:

$$\mathcal{D}_{1/N}(M) = \exp \left[i \int \left(\frac{\pi}{N} \star j + \frac{N}{4\pi} a \wedge da + \frac{1}{2\pi} a \wedge dA \right) \right]$$

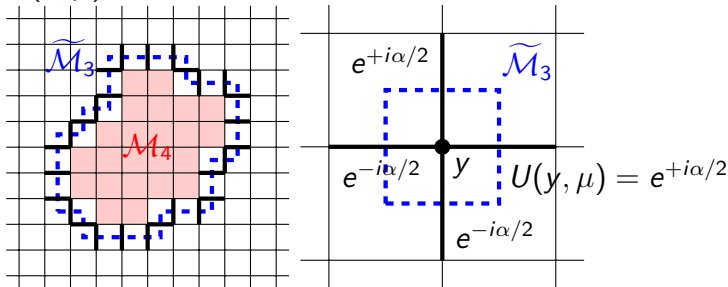
- Gauge invariant and topological; no inverse operator

$$\mathcal{D}_{1/N} \times \mathcal{D}_{1/N}^\dagger = \mathcal{C} \neq 1$$

- ▶ $\mathcal{C}$: condensation operator

Introduction of topological defect

- Construct (fermion measure in) **anomalous** chiral gauge theory
 - Introduce external $U(1)'$ gauge group
 - to manipulate axial rotation (Chiral $U(1)' \ni U(x, \mu)$)
 - to define topological defect
 - Keep locality for **dynamical** fields (Vector $U(1) \ni u(x, \mu)$)
- $U(x, \mu) = e^{\pm i\alpha/2}$ for 3D closed surface $\widetilde{\mathcal{M}}_3$; otherwise 1



- Symmetry operator may be written by

$$\langle U_\alpha(\widetilde{\mathcal{M}}_3) \mathcal{O} \rangle_F := \langle \mathcal{O} \rangle_F^{\widetilde{\mathcal{M}}_3} \exp \left[i\alpha\gamma \sum_{x \in \mathcal{M}_4} \epsilon_{\mu\nu\rho\sigma} f_{\mu\nu}(x) f_{\rho\sigma}(x + \hat{\mu} + \hat{\nu}) \right]$$

Coupling with $\mathbb{Z}_N$ TQFT for $\alpha \sim 2\pi/N$

- Set rotation angles α as $2\pi p/N$ ($p, N \in \mathbb{Z}$)
- It is natural to consider 3D level- N BF theory on $\mathcal{M}_3$

$$S_{\text{BF}} = -\frac{ip\pi}{N} \sum_{x \in \mathcal{M}_3} \epsilon_{\mu\nu\rho} \left\{ b_\mu(\tilde{x}) \left[\partial_\nu c(x + \hat{\mu}) - \frac{1}{2} z_{\nu\rho}(x + \hat{\mu}) \right] - \frac{1}{2} z_{\mu\nu}(x) c_\rho(x + \hat{\mu} + \hat{\nu}) \right\}$$

- ▶ Dual lattice $\tilde{x}$; $f = \delta a + 2\pi z$, $-\pi < a_\mu(x) \equiv \frac{1}{i} \ln u(x, \mu) \leq \pi$
 - ▶ For simplicity, $S_{\text{BF}} = \frac{-ip\pi}{N} \sum [b(\delta c - z) - z \cup c]$
- Consider $\mathcal{Z}_{\mathcal{M}_3}[z] = \frac{1}{N^s} \int DbDc e^{-S_{\text{BF}}}$, s : # of sites $\in \mathcal{M}_3$
- From summation over b , $\mathcal{Z}_{\mathcal{M}_3}[z] = 0$ if $\sum_{\mathcal{M}_2} z \neq 0 \bmod N$
 - ▶ If $z = 0$, $\mathcal{Z}_{\mathcal{M}_3}[0] = N^{b_2-1}$, where b_2 : 2nd Betti number of $\mathcal{M}_3$
 - ▶ If $z = \delta\nu \bmod N$,

$$\mathcal{Z}_{\mathcal{M}_3}[z] = \exp \left(-\frac{ip\pi}{N} \sum_{\mathcal{M}_3} \delta\nu \cup \nu \right) \mathcal{Z}_{\mathcal{M}_3}[0]$$

Symmetry operator in terms of $\mathbb{Z}_N$ TQFT

- Redefine symmetry operator on arbitrary 3-cycle $\widetilde{\mathcal{M}}_3$:

$$\left\langle \tilde{U}_{\frac{2\pi p}{N}}(\widetilde{\mathcal{M}}_3) \mathcal{O} \right\rangle_F = \langle \mathcal{O} \rangle_F^{\widetilde{\mathcal{M}}_3} \exp \left[-\frac{ip}{4\pi N} \sum_{\mathcal{M}_3} (a \cup f + 2\pi z \cup a) \right] \mathcal{Z}_{\mathcal{M}_3}[z]$$

- ▶ Then, topological: $\left\langle \tilde{U}_{\frac{2\pi p}{N}}(\widetilde{\mathcal{M}}'_3) \mathcal{O} \right\rangle_F = \left\langle \tilde{U}_{\frac{2\pi p}{N}}(\widetilde{\mathcal{M}}_3) \mathcal{O} \right\rangle_F$
- ▶ Gauge invariant: cancellation between exp and $\mathcal{Z}_{\mathcal{M}_3}$ under $a \mapsto a - \delta\phi - 2\pi l$ and $z \mapsto z + \delta l$

- $\mathcal{Z}_{\mathcal{M}_3}^{(p,N)}[z] \mathcal{Z}_{-\mathcal{M}_3}^{(p,N)}[z] = \mathcal{Z}_{\mathcal{M}_3}[0] \mathcal{C}_{\mathcal{M}_3}[z]$; condensation operator:

$$\mathcal{C}_{\mathcal{M}_3}[z] = \frac{1}{N^s} \int Db Dc e^{b(\delta c - z)}$$

- For different p_1 and p_2 (assuming $\gcd(p_1 + p_2, N) = 1$)

$$\tilde{U}_{2\pi p_1/N} \tilde{U}_{2\pi p_2/N} = \mathcal{Z}_{\mathcal{M}_3}[0] \tilde{U}_{2\pi(p_1+p_2)/N}$$

- ▶ More generally

$$\tilde{U}_{2\pi p_1/N_1} \tilde{U}_{2\pi p_2/N_2} = \frac{\mathcal{Z}_{\mathcal{M}_3}^{(N_1)}[0] \mathcal{Z}_{\mathcal{M}_3}^{(N_2)}[0]}{\mathcal{Z}_{\mathcal{M}_3}^{(N)}[0]} \tilde{U}_{2\pi(p_1/N_1 + p_2/N_2)}$$

Summary

- Generalized symmetries have been developed in this decade
- Standing on a **fully regularized framework: lattice gauge theory**
 - ▶ Fractional topological charge, etc.
 - ▶ Numerical simulation of $SU(N)$ gauge theory coupled with B_p
[Abe–OM–Suzuki, Kanamori–OM–Tanizaki–Tomiya–Watanabe]
- Again standing on it, consider non-invertible symmetry
 - ▶ Construction of fermion measure: for dynamical fields, smooth and local; for external $U(x, \mu)$, **non-local** (unphysical)
 - ▶ Symmetry operator for rational angles and evaluation of **fusion rules!**
 - ▶ Questions
 - ★ Generalization to non-Abelian gauge theory
 - ★ Some aspects of anomalous/chiral LGT (w/ boundary)
- (Above is given on Wilson lattice; how about Villain lattice?)

$U(1) \times U(1)'$ lattice gauge theory: Gauge

- Vector $U(1) \ni u(x, \mu)$ (physical) and Chiral $U(1)' \ni U(x, \mu)$ (non-dynamical) gauge group
- Expectation value with magnetic flux m :

$$\langle \mathcal{O} \rangle = \frac{1}{Z} \int \mathrm{D}u \, e^{-S_G} \langle \mathcal{O} \rangle_F, \quad \langle \mathcal{O} \rangle_F = w[m] \int \mathrm{D}\psi \mathrm{D}\bar{\psi} \, e^{-S_F} \mathcal{O}.$$

- (Unconventional) Gauge action S_G following [Lüscher],

$$S_G = \frac{1}{4g_0^2} \sum_{x \in \Gamma} \sum_{\mu, \nu} \mathcal{L}_{\mu\nu}(x)$$
$$\mathcal{L}_{\mu\nu}(x) = \begin{cases} [f_{\mu\nu}(x)]^2 \left\{ 1 - \frac{[f_{\mu\nu}(x)]^2}{\epsilon^2} \right\}^{-1} & \text{if } |f_{\mu\nu}(x)| < \epsilon, \\ \infty & \text{otherwise.} \end{cases}$$

for $0 < \epsilon < \pi/3$, where the field strength of $u(x, \mu)$ is

$$f_{\mu\nu}(x) = \frac{1}{i} \ln u(x, \mu) u(x + \hat{\mu}, \nu) u(x + \hat{\nu}, \mu)^{-1} u(x, \nu)^{-1}.$$

$U(1) \times U(1)'$ lattice gauge theory: Fermion

- Two left-handed Weyl fermions $\psi_{1,2}$ possess $U(1)$ charges

$$(e_1, e'_1) = (+1, -1), \quad (e_2, e'_2) = (-1, -1).$$

- Since overlap Dirac operator fulfills Ginsparg–Wilson relation

$$\gamma_5 D + D \gamma_5 = D \gamma_5 D,$$

chirality projection operators are defined by

$$\hat{\gamma}_5 = \gamma_5(1 - D), \quad \hat{P}_{\pm} = \frac{1}{2}(1 \pm \hat{\gamma}_5), \quad P_{\pm} = \frac{1}{2}(1 \pm \gamma_5).$$

- Fermion action is

$$S_F = \sum_{x \in \Gamma} \bar{\psi}(x) D \psi(x) = \sum_{x \in \Gamma} \bar{\psi}(x) P_+ D \hat{P}_- \psi(x)$$

- Fermion integration measure is

$$D\psi D\bar{\psi} = \prod_j dc_j \prod_k d\bar{c}_k, \quad \psi(x) = \sum_j v_j(x) c_j, \quad \bar{\psi}(x) = \sum_k \bar{c}_k \bar{v}_k(x)$$

Basis $v_j(x)$ are in projected space $\hat{P}_- v_j = v_j$, $(v_k, v_j) = \delta_{kj}$

Construction of fermion integration measure

- Basis vectors depend on gauge field due to $\hat{P}_-$
- Under variation δ of gauge field, δv_j is not determined. . .
 - ▶ Requirements: (single-valued, gauge-inv) smooth, & local
- Under infinitesimal variations $\delta_\eta u(x, \mu) = i\eta_\mu(x)u(x, \mu)$,
 $\delta_\eta U(x, \mu) = i\eta'_\mu(x)U(x, \mu)$,

$$\delta_\eta \ln \det \bar{v} D v = (\text{term from } \delta_\eta D) + (\text{term from } \delta_\eta v).$$

Second term is called measure term (measure currents)

$$-i\mathcal{L}_\eta = \sum_j (v_j, \delta_\eta v_j) := -i \sum_{x \in \Gamma} [\eta_\mu(x) j_\mu(x) + \eta'_\mu(x) J_\mu(x)] .$$

- ▶ Consistent/covariant anomaly: $\mathcal{A}_{\text{cons}} = \frac{1}{3} \mathcal{A}_{\text{cov}}$ for Abelian GT
- ▶ To remedy this issue, we need to include counterterm [E.g., see Fujikawa–Suzuki Chap. 6]

Construction of measure currents

- For $\eta_\mu(x) = -\partial_\mu \omega(x)$, $\eta'_\mu(x) = -\partial_\mu \Omega(x)$, ($\gamma = -\frac{1}{32\pi^2}$)

$$\delta_\eta \langle \mathcal{O} \rangle_F = \langle \delta_\eta \mathcal{O} \rangle_F - 2i\gamma \sum_{x \in \Gamma} \Omega(x) \epsilon_{\mu\nu\rho\sigma} f_{\mu\nu}(x) f_{\rho\sigma}(x + \hat{\mu} + \hat{\nu}) \langle \mathcal{O} \rangle_F. \quad (\star)$$

- To prove this, measure currents fulfill the following conditions:
 - ① depend smoothly and locally on gauge fields
 - ② satisfy “integrability condition” ($[\delta_\eta, \delta_\zeta]$)
 - ③ satisfy “anomalous conservation law” ($\partial^* j, \partial^* J$)
- We find **non-local** dependence of $U(x, \mu)$; but $U(x, \mu)$ is external
- In infinite volume, one can construct the currents explicitly w.r.t.

$$u(x, \mu) = e^{i\mathfrak{a}_\mu(x)}, \quad |\mathfrak{a}_\mu(x)| \leq \pi(1 + 8\|x\|), \quad f(x) = d\mathfrak{a}(x),$$
$$U(x, \mu) = e^{i\mathfrak{A}_\mu(x)}, \quad |\mathfrak{A}_\mu(x)| \leq \pi(1 + 8\|x\|), \quad F(x) = d\mathfrak{A}(x),$$

as follows:

Backup: Construction of measure currents $L \rightarrow \infty$

$$\begin{aligned}
 \mathfrak{L}_\eta^{\star\text{inv}} &= i \int_0^1 ds \operatorname{Tr}(\hat{P}_- [\partial_s \hat{P}_-, \delta_\eta \hat{P}_-]) + \int_0^1 ds \sum_{x \in \mathbb{Z}^4} [\eta_\mu(x) k_\mu(x) + \mathfrak{a}_\mu(x) \delta_\eta k_\mu(x)] \\
 &\quad + \int_0^1 ds \sum_{x \in \mathbb{Z}^4} [\eta'_\mu(x) K_\mu(x) + \mathfrak{A}_\mu(x) \delta_\eta K_\mu(x)] \\
 &\quad - \frac{4}{3} \gamma \sum_{x \in \mathbb{Z}^4} \epsilon_{\mu\nu\rho\sigma} \{ \eta'_\mu(x) \mathfrak{a}_\nu(x + \hat{\mu}) f_{\rho\sigma}(x + \hat{\mu} + \hat{\nu}) \\
 &\quad \quad \quad + \mathfrak{A}_\mu(x) \delta_\eta [\mathfrak{a}_\nu(x + \hat{\mu}) f_{\rho\sigma}(x + \hat{\mu} + \hat{\nu})] \} \\
 &:= \sum_{x \in \mathbb{Z}^4} [\eta_\mu(x) j_\mu^{\star\text{inv}}(x) + \eta'_\mu(x) J_\mu^{\star\text{inv}}(x)] , \\
 \mathfrak{L}_\eta^{\star\text{non-inv}} &= 4\gamma \sum_{x \in \mathbb{Z}^4} \epsilon_{\mu\nu\rho\sigma} \{ \eta'_\mu(x) \mathfrak{a}_\nu(x + \hat{\mu}) f_{\rho\sigma}(x + \hat{\mu} + \hat{\nu}) \\
 &\quad \quad \quad + \mathfrak{A}_\mu(x) \delta_\eta [\mathfrak{a}_\nu(x + \hat{\mu}) f_{\rho\sigma}(x + \hat{\mu} + \hat{\nu})] \} \\
 &:= \sum_{x \in \mathbb{Z}^4} [\eta_\mu(x) j_\mu^{\star\text{non-inv}}(x) + \eta'_\mu(x) J_\mu^{\star\text{non-inv}}(x)] .
 \end{aligned}$$